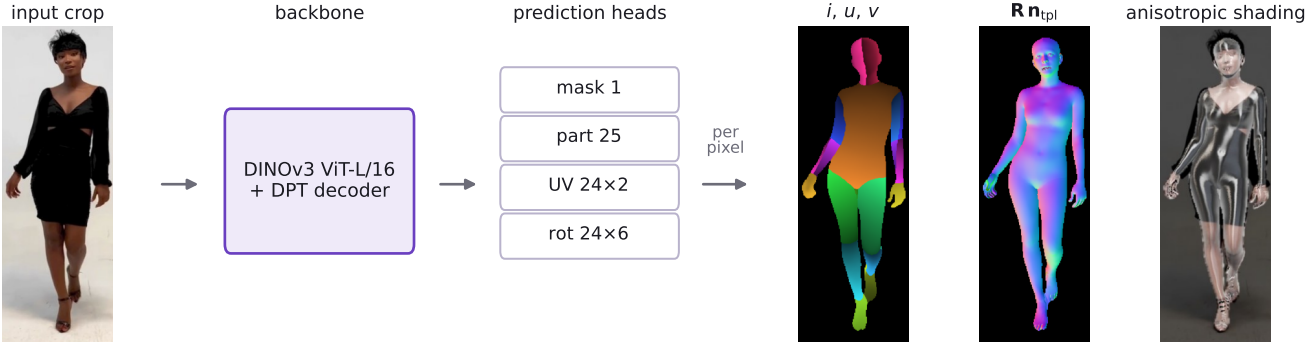


per pixel: body part i, chart coordinates (u, v), and a full frame $\mathbf{R} \in SO(3)$ — supervised with the CSC loss $\mathcal{T}(i, u, v) \to \mathbf{X} \in \mathbb{R}^3$